

6.3 Bisector of an angle in a triangle

Theorem 18

The bisector of an angle in a triangle divides the opposite side in the ratio of the sides containing the angle.

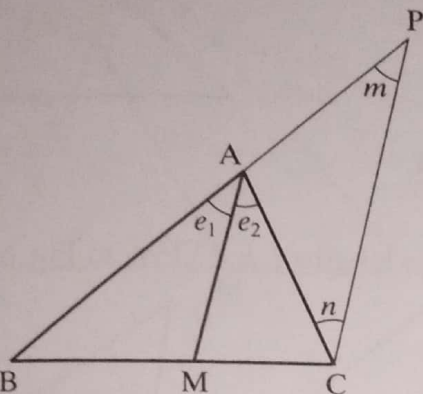


Fig. 6.17

Given: $\triangle ABC$ with $\overline{AM}$ bisecting $\widehat{A}$ and meeting $\overline{BC}$ at M .

To prove: $\frac{|BA|}{|AC|} = \frac{|BM|}{|MC|}$

Construction: Draw $\overline{CP} \parallel \overline{MA}$, to cut $\overline{BA}$ produced at P .

Proof:

With the lettering used in Fig. 6.17,

$$\begin{aligned} m &= e_1 && (\text{corr. angles}) \\ &= e_2 && (\text{given } e_1 = e_2) \\ &= n && (\text{alt. angles}) \end{aligned}$$

$\therefore \triangle PAC$ is isosceles with $|AC| = |AP|$.

$$\therefore \frac{|BA|}{|AC|} = \frac{|BA|}{|AP|}$$

$$\therefore \frac{|BA|}{|AC|} = \frac{|BM|}{|MC|} \quad (\overline{AM} \parallel \overline{PC})$$

Example 4

In $\triangle XYZ$, $|XY| = 7$ cm, $|YZ| = 5$ cm, $|XZ| = 10$ cm and the bisector of $\widehat{Y}$ meets $\overline{XZ}$ at P . Calculate $|XP|$ and $|PZ|$.

Fig. 6.18 is a sketch of the data in the question.

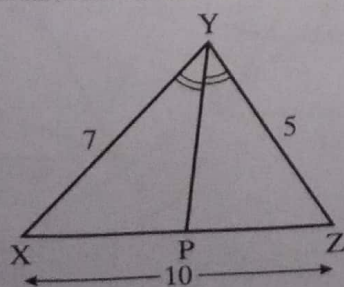


Fig. 6.18

$$\frac{|XY|}{|YZ|} = \frac{|XP|}{|PZ|}$$

$$\therefore \frac{|XP|}{|PZ|} = \frac{7}{5}$$

$$\begin{aligned} \therefore |XP| &= \frac{7}{12} \text{ of } |XZ| \\ &= \frac{7}{12} \text{ of } 10 \text{ cm} \\ &= \frac{70}{12} \text{ cm} = \frac{35}{6} \text{ cm} = 5\frac{5}{6} \text{ cm} \\ \therefore |PZ| &= 10 \text{ cm} - 5\frac{5}{6} \text{ cm} \\ &= 4\frac{1}{6} \text{ cm} \end{aligned}$$

Exercise 6d

- ① Find **i** the ratio $x:y$, **ii** the values of x and y , in each part of Fig. 6.19.

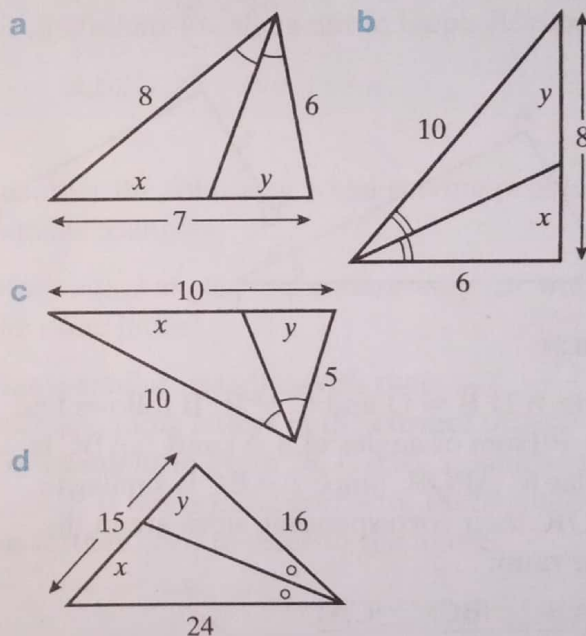


Fig. 6.19

- ② In $\triangle ABC$, $|AB| = 7$ cm, $|BC| = 9$ cm, $|CA| = 5$ cm and the bisector of $\widehat{A}$ meets $\overline{BC}$ at D . Calculate $|BD|$ and $|DC|$.
- ③ In $\triangle PQR$, $|PQ| = 15$ cm, $|QR| = 16$ cm, $|RP| = 9$ cm. The bisector of $\widehat{P}$ meets $\overline{QR}$ at A and M is the mid-point of $\overline{QR}$. Calculate $|AM|$.
- ④ In $\triangle ABC$, $|AB| = 12$ cm, $|BC| = 14$ cm, $|CA| = 9$ cm. X is a point on $\overline{BC}$ such that $|BX| = 8$ cm. Show that $\overline{AX}$ bisects $\widehat{A}$.

- 5 In Fig. 6.20 calculate which of the points X and Y is closer to A and by how much.

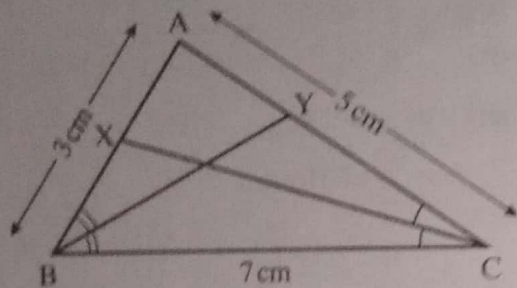


Fig. 6.20

6.4 Similar triangles

Similar triangles have the same shape but are of different size. JSS Book 3 showed that two triangles are similar if the angles of one are respectively equal to the angles of the other.

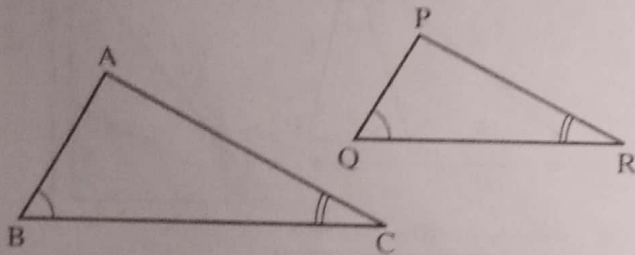


Fig. 6.21

In Fig. 6.21 $\widehat{B} = \widehat{Q}$ and $\widehat{C} = \widehat{R}$. It follows that $\widehat{A} = \widehat{P}$ (sum of angles of a $\triangle$) and $\triangle ABC$ is similar to $\triangle PQR$. Since $\triangle ABC$ is similar to $\triangle PQR$, their corresponding sides are in the same ratio:

$$\frac{|AB|}{|PQ|} = \frac{|BC|}{|QR|} = \frac{|CA|}{|RP|}$$

Exercise 6e (Revision)

- 1 Copy each part of Fig. 6.22. Mark in the pairs of equal angles. Hence find the ratio $x:y$ in each figure.

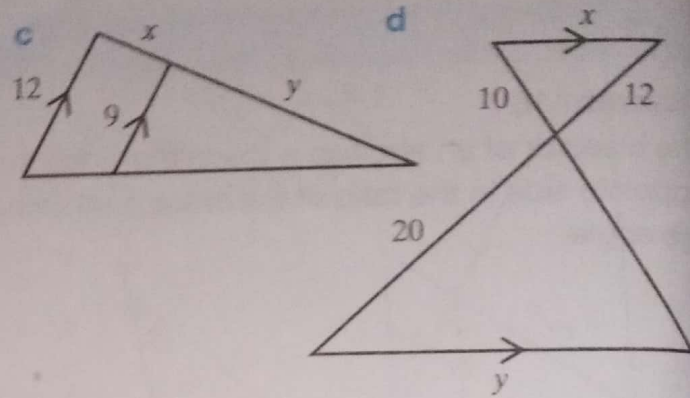
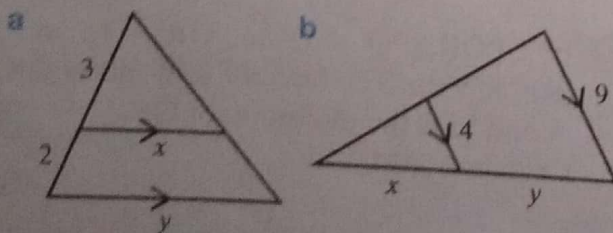


Fig. 6.22

- 2 Find the lengths $j, k, l, \dots, w$ in Fig. 6.23.

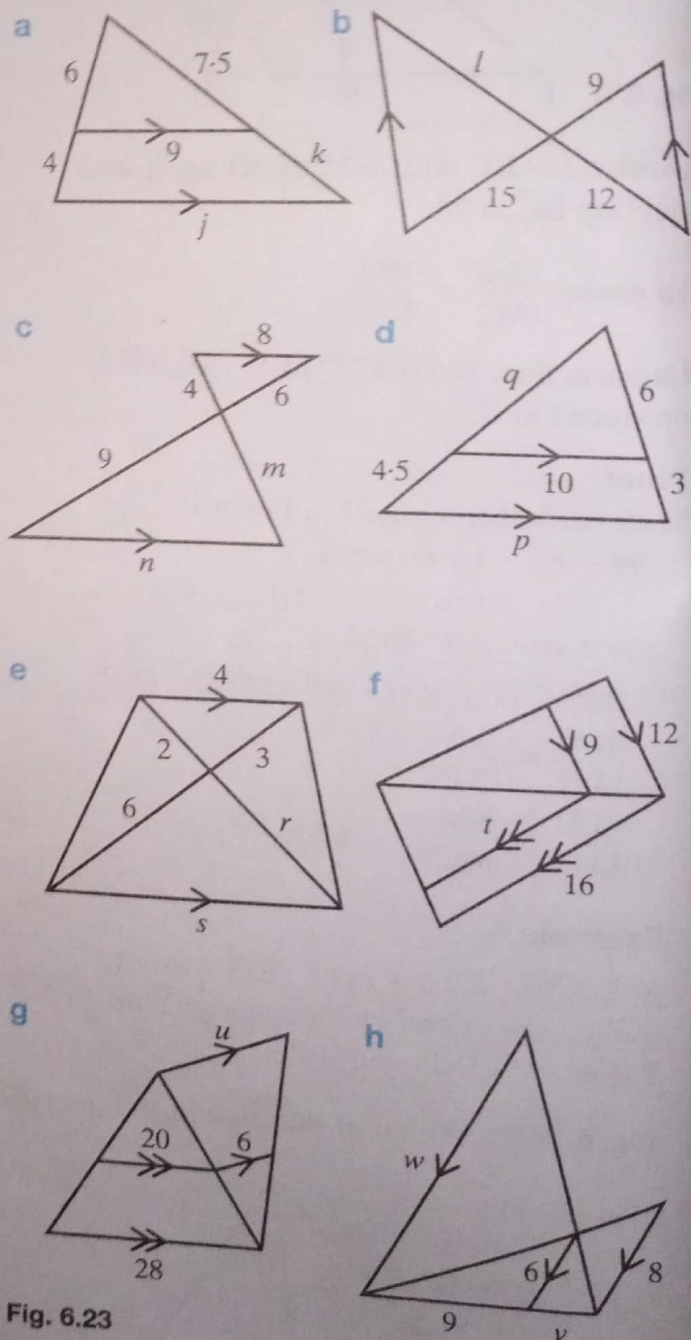


Fig. 6.23

Example 5

Given Fig. 6.24 with $|QL| = 2$ cm, $|LM| = 3$ cm and $|PR| = 4$ cm, calculate $|QR|$. [WAEC]

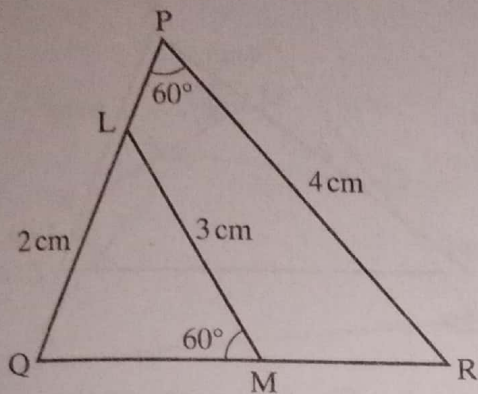


Fig. 6.24

In $\triangle s$ MQL and PQR,

$$\widehat{LMQ} = \widehat{RPQ} \quad (\text{given } 60^\circ)$$

$$\widehat{MQL} = \widehat{PQR} \quad (\text{same angle})$$

$$\therefore \widehat{QLM} = \widehat{QRP} \quad (\text{sum of angles of } \triangle)$$

$\therefore \triangle MQL$ is similar to $\triangle PQR$.

$$\therefore \frac{|MQ|}{|PQ|} = \frac{|QL|}{|QR|} = \frac{|LM|}{|RP|}$$

Substituting the given values:

$$\frac{|MQ|}{|PQ|} = \frac{2}{|QR|} = \frac{3}{4}$$

The 2nd and 3rd ratios will give the required result.

$$\frac{2}{|QR|} = \frac{3}{4} \quad \text{or} \quad \frac{|QR|}{2} = \frac{4}{3}$$

$$|QR| = \frac{2 \times 4}{3} = \frac{8}{3} = 2\frac{2}{3} \text{ cm}$$

Example 6

In Fig. 6.25, show that $\triangle XBC$ is similar to $\triangle XDA$. Hence find $|AB|$ if $|XA| = 9$ cm, $|XC| = 10$ cm and $|CD| = 8$ cm.

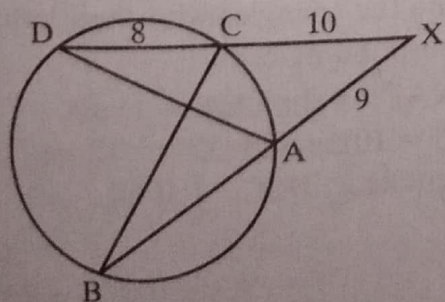


Fig. 6.25

In $\triangle s$ XBC and XDA,

$$\widehat{BXC} = \widehat{DXA} \quad (\text{same angle})$$

$$\widehat{XBC} = \widehat{XDA} \quad (\text{angles in same segment})$$

$$\therefore \widehat{XCB} = \widehat{XAD} \quad (\text{sum of angles of } \triangle)$$

$\therefore \triangle XBC$ is similar to $\triangle XDA$.

$$\therefore \frac{|XB|}{|XD|} = \frac{|BC|}{|DA|} = \frac{|CX|}{|AX|}$$

Substituting the given lengths:

$$\frac{|AB| + 9}{10 + 8} = \frac{|BC|}{|DA|} = \frac{10}{9}$$

The 1st and 3rd ratios will give the required result.

$$\frac{|AB| + 9}{18} = \frac{10}{9}$$

$$|AB| + 9 = \frac{18 \times 10}{9} = 20$$

$$|AB| = 20 - 9 = 11 \text{ cm}$$

Remember the following when solving problems on similar triangles:

- 1 Mark equal angles in the same way (or with the same letter).
- 2 Name similar triangles with their corresponding letters in the correct order. For example, in Fig. 6.26, $\triangle ABC$ is similar to $\triangle AQP$. If this is done correctly, there is then no further need to refer to the figure.

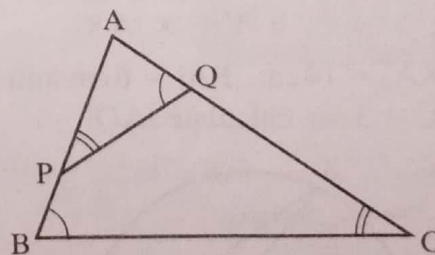


Fig. 6.26

- 3 Write down *all three* ratios, whether or not they are all required for the answer. For example, in Fig. 6.26.

$$\frac{|AB|}{|AQ|} = \frac{|BC|}{|QP|} = \frac{|CA|}{|PA|}$$

- 4 Substitute the given values into the ratios, then choose a pair of ratios that lead to the required result.

Exercise 6f

- ① Calculate p, q, r, s, t in Fig. 6.27.

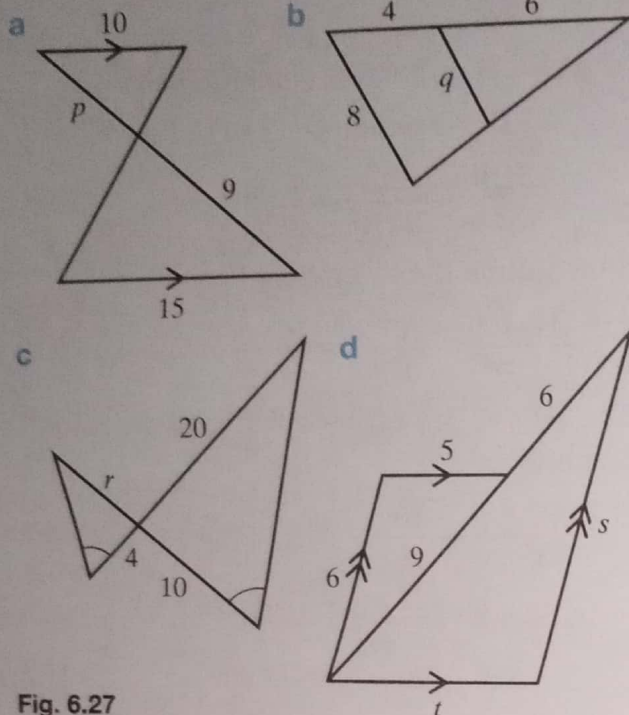


Fig. 6.27

- ② In Fig. 6.26, calculate $|AB|$ if $|AQ| = 2$ cm, $|BC| = 11$ cm and $|QP| = 6$ cm.
- ③ In Fig. 6.26, calculate $|QC|$ if $|AP| = 7$ cm, $|PB| = 3$ cm and $|AQ| = 5$ cm.
- ④ a In Fig. 6.28, show that $\triangle PAX$ is similar to $\triangle YAQ$.
 b Hence show that $|PA| \times |AQ| = |YA| \times |AX|$.
 c If $|XA| = 14$ cm, $|PA| = 6$ cm and $|YA| = 3$ cm, calculate $|AQ|$.

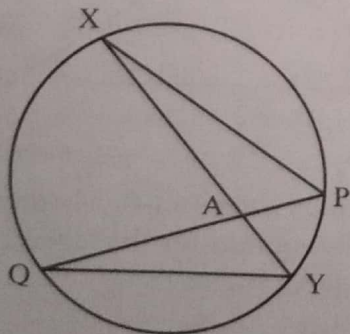


Fig. 6.28

- ⑤ In Fig. 6.29, $|QX| = 4$ cm, $|XR| = 6$ cm, $|RZ| = 2$ cm and $\angle XQR = \angle ZYR$. Calculate $|YQ|$. [WAEC]

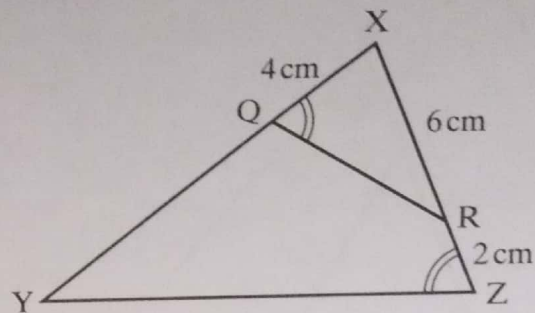


Fig. 6.29

- ⑥ a Show that Fig. 6.30 contains three similar triangles.

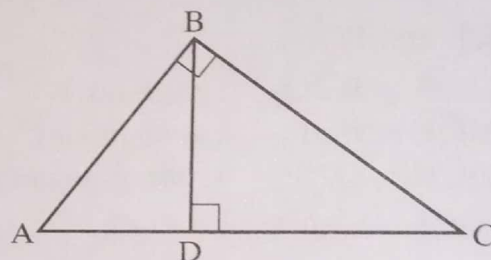


Fig. 6.30

- b Name the triangles, with their letters in the correct corresponding order.
- c Hence name two ratios equal to $\frac{|AD|}{|AB|}$.
- ⑦ a Show that Fig. 6.31 contains a pair of similar triangles.

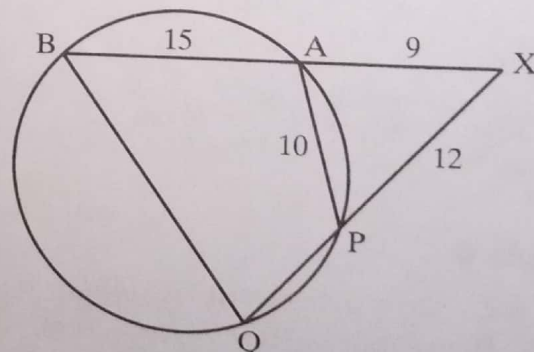


Fig. 6.31

- b Name the triangles with their letters in corresponding order.
- c If $|XA| = 9$ cm, $|AB| = 15$ cm, $|AP| = 10$ cm and $|XP| = 12$ cm. Calculate i $|BQ|$, ii $|PQ|$.

- 8 In Fig. 6.32, $\widehat{PQX} = \widehat{XZY}$, $|PQ| = 2$ cm, $|XP| = 3$ cm, $|XQ| = 4$ cm and $|QY| = 2$ cm. Calculate $|YZ|$. [WAEC]

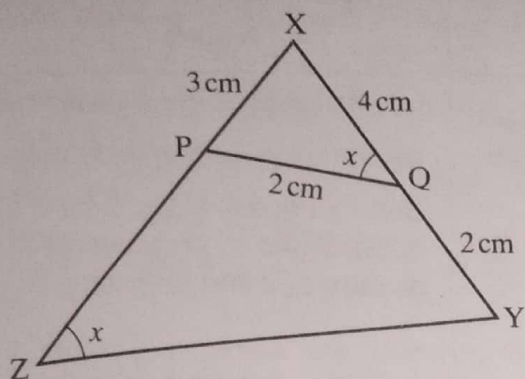


Fig. 6.32

- 9 In Fig. 6.33, O is the centre of the circle and $\widehat{AOX} = 90^\circ$. If $|AX| = 8$ cm and $|XY| = 1$ cm, calculate the radius of the circle.

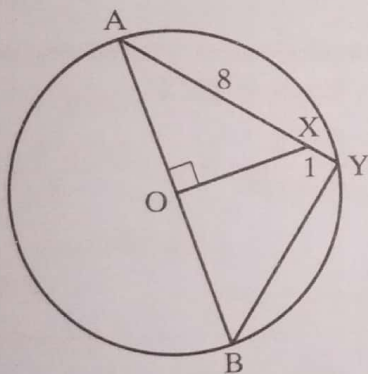


Fig. 6.33

- 10 Calculate the lengths marked with letters in Fig. 6.34.

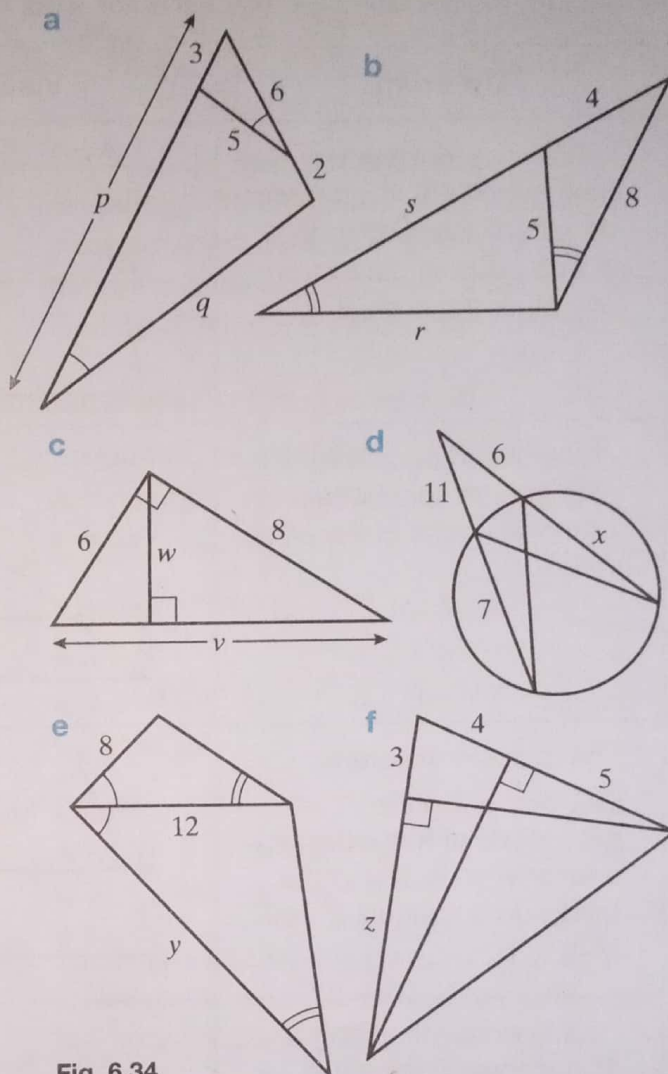
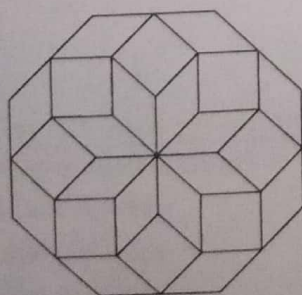


Fig. 6.34

Challenge Box

Regular Polygons

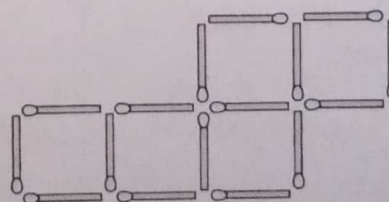
Can you see any regular polygons in this pattern?



How many of each kind?

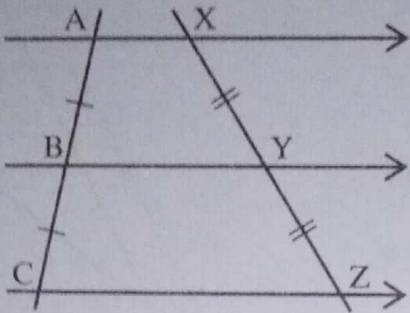
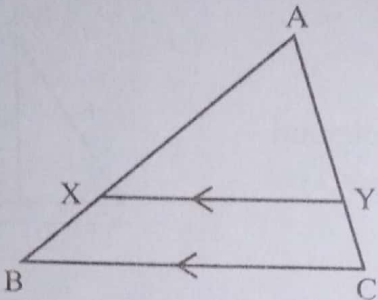
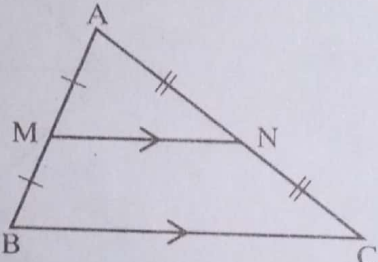
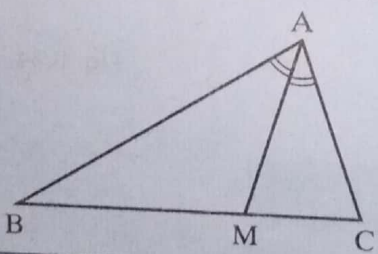
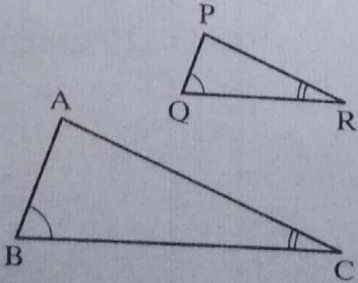
Challenge Box

Changing positions



Change the positions of just two matchsticks to reduce the number of squares from five to four.

Remember the following geometrical ratios when calculating lengths.

Theorem	Visual aid	Notes
1 If three or more parallel lines cut off equal intercepts on a transversal, then they cut off equal intercepts on any other transversal.		If $AB = BC$ then $XY = YZ$ See Examples 1 and 2 for applications to proportional division of a line segment.
2 A line drawn parallel to one side of a triangle divides the other two sides in the same ratio.		$\frac{ AX }{ XB } = \frac{ AY }{ YC }$ See Example 3.
3 The midpoint theorem The line joining the mid-points of two sides of a triangle is parallel to the third side and equal to half of it.		$ MN = \frac{1}{2} BC $ and $\overline{MN} \parallel \overline{BC}$
4 The bisector of an angle in a triangle divides the opposite side in the ratio of the sides containing the angle.		$\frac{ AB }{ AC } = \frac{ MB }{ MC }$ See Example 4.
5 Similar triangles are equiangular and have corresponding sides in the same ratio.		$\frac{ AB }{ PQ } = \frac{ BC }{ QR } = \frac{ CA }{ RP }$ See Examples 5 and 6.

Revision exercises and tests

Chapters 1–6

Revision exercise 1 (Chapters 1, 4)

- ① Write the following numbers in standard form.
a 5 000 **b** 0.038
c 75.48 **d** 0.000 025
- ② Use tables to calculate the following.
a $391 \times 7\,925$ **b** $465.8 \div 42.55$
c $\frac{56 \times 714}{116}$ **d** $\frac{9}{7.806}$
- ③ Use tables to calculate the following.
a $\left(\frac{841.4}{14.8}\right)^3$ **b** $\sqrt[4]{631\,300}$
- ④ Use tables to calculate the following.
a $163 \times 0.002\,07$ **b** $0.770\,8 \div 0.000\,39$
c $\frac{0.56 \times 0.071\,4}{1\,160}$ **d** $\frac{9}{780.6}$
- ⑤ Calculate the following.
a $\frac{(0.538)^2}{2.655}$ **b** $\sqrt[3]{0.080\,05}$
- ⑥ Use square root and reciprocal tables to find the values of the following correct to 2 s.f.
a $\frac{1}{\sqrt{910}}$ **b** $\frac{\sqrt{99.8}}{6.8}$
- ⑦ Taking π to be 3.14, calculate $\pi(7.6)^2$ to an appropriate degree of accuracy.
- ⑧ Calculate the following, giving the answers to 2 s.f. (Use $\pi = 3.14$)
a $\sqrt{(0.36)^2 + (0.54)^2}$
b $\frac{4}{3}\pi(2.5)^3$
- ⑨ Calculate the possible percentage errors in the following measurements if each is given to 1 significant figure.
a a door 200 cm high
b a book 8 mm thick
c a box of mass 30 kg
d a farm of area 500 ha

- ⑩ The floor of a rectangular classroom is measured as 7.2 m by 6.8 m to 2 s.f. Use these measurements to calculate to 1 d.p. the maximum and minimum areas of the floor in m^2 . Find the difference between your two answers.

Revision test 1 (Chapters 1, 4)

- ① The antilog of $\bar{2}.3869$ is
A 0.002 437 **B** 0.024 37 **C** 0.243 7
D 2.437 **E** 24.37
- ② Express 0.007 8 in standard form.
A 7.8×10^{-3} **B** 7.8×10^{-2}
C 7.8×10^1 **D** 7.8×10^2
E 7.8×10^3
- ③ Express 4081.039 2 to 2 significant figures.
A 4 100 **B** 4081.04 **C** 4 081.039
D 4080 **E** 4000
- ④ The length of a wall is given as 20 m to 1 s.f. Calculate the possible percentage error in this measurement.
A $\pm 0.25\%$ **B** $\pm 2.5\%$ **C** $\pm 5\%$
D $\pm 10\%$ **E** $\pm 25\%$
- ⑤ Estimate which one of the following has the least value.
A $\frac{0.256}{48}$ **B** $\frac{0.256}{4.8}$ **C** $\frac{2.56}{0.48}$
D $\frac{2.56}{4.8}$ **E** $\frac{25.6}{480}$
- ⑥ Use tables to calculate **a** $0.650\,2^4$, **b** $\sqrt[4]{0.650\,2}$, giving each answer correct to 2 s.f.
- ⑦ Express 36.876 1 correct to
a 2 d.p., **b** 2 s.f., **c** 3 s.f.,
d the nearest whole number.
- ⑧ A town has a population density of 1 100 people per km^2 . If the town covers an area of about 18 km^2 , estimate its population to 1 s.f.

- ⑨ If $\pi r^2 = 270$, calculate the value of r to a suitable degree of accuracy. Take π to be 3.1.
- ⑩ The side of an equilateral triangle is measured as 7 cm to the nearest centimetre. What are the greatest and least possible lengths of the side of the triangle? Use these values to calculate the greatest and least possible values of the area of the triangle, giving your answers to the nearest 0.1 cm².

Revision exercise 2 (Chapters 2, 6)

- ① Each angle of a regular polygon is 162° . How many sides does the polygon have?
- ② In $\triangle ABC$, $\widehat{BAC} = 68^\circ$ and $\widehat{ABC} = 30^\circ$. BC is produced to X . The bisectors of $\widehat{ABC}$ and $\widehat{ACX}$ meet at P . Calculate $\widehat{BCP}$ and $\widehat{BPC}$.
- ③ $ABCD$ is a quadrilateral in which $|AB| = |CD|$ and $|AD| = |BC|$. Prove that $\overline{AD}$ is parallel to $\overline{BC}$. (Hint: Join $\overline{AC}$ or $\overline{BD}$.)
- ④ In Fig. R1, $\widehat{ABP} = 110^\circ$ and $\widehat{DCP} = 163^\circ$. Calculate $\widehat{BPC}$.

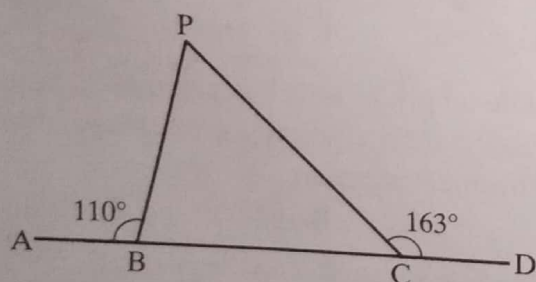


Fig. R1

- ⑤ Use ruler, compasses and set square only.
- a Draw a line $\overline{AB}$ 7 cm long. Construct a point P on $\overline{AB}$ such that $|AP| : |PB| = 1 : 3$. Measure $|PB|$.
- b Draw a line 8 cm long. Divide the line into six equal parts. Hence find $\frac{5}{6}$ of 8 cm.
- ⑥ In Fig. R2, O is the centre of the circle. Find the value of x .

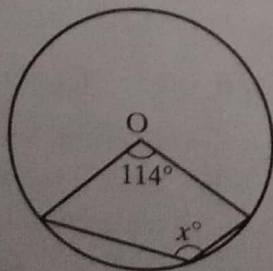


Fig. R2

- ⑦ In Fig. R3, $\overline{AB}$ is a diameter of the circle. Calculate the sizes of the following angles.
- a $\widehat{ABP}$ b $\widehat{ABZ}$ c $\widehat{AYZ}$
d $\widehat{BYZ}$ e $\widehat{ABY}$ f $\widehat{YZB}$

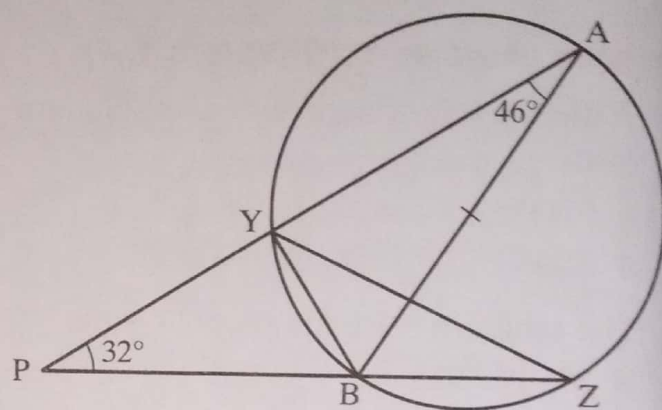


Fig. R3

- ⑧ In Fig. R4, X, Y, Z are the mid-points of $\overline{AB}$, $\overline{BC}$, $\overline{CA}$ respectively. Calculate the lengths of the sides of $\triangle XYZ$.

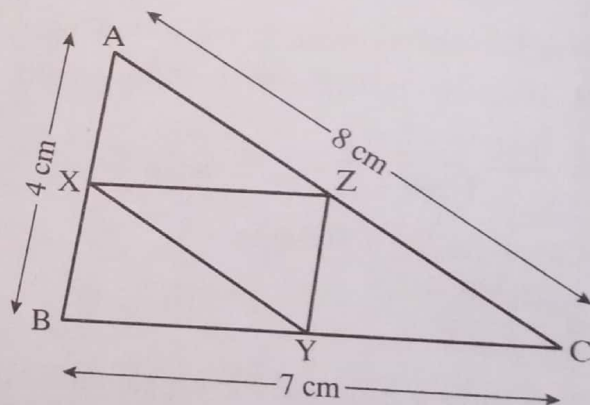


Fig. R4

- ⑨ L, M, N, P are points on the circumference of a circle, centre O . $\overline{LN}$ and $\overline{PM}$ intersect at X . $\widehat{NLM} = 35^\circ$ and $\widehat{LXP} = 98^\circ$. Calculate a $\widehat{LNP}$, b $\widehat{LOP}$.
- ⑩ In Fig. R5, prove that $\widehat{RPQ} + \widehat{PQS} = 180^\circ$. What can be said about the lines $\overline{RP}$ and $\overline{SQ}$?

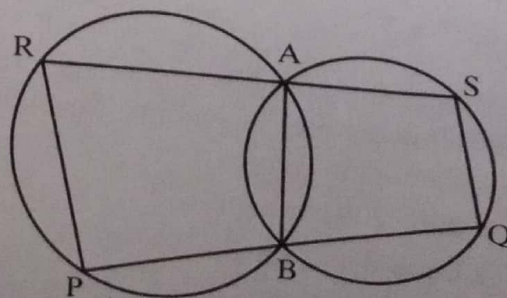


Fig. R5

Revision test 2 (Chapters 2, 6)

- ① The angles of a pentagon are x° , $2x^\circ$, $(x + 30)^\circ$, $(x - 10)^\circ$ and $(x + 40)^\circ$. $x =$
A 30 **B** 50 **C** 60 **D** 80 **E** 108

- ② ABCD is a quadrilateral such that $\widehat{ABC} = \widehat{ADC} = 90^\circ$ and $|AD| = |BC|$. Which one of the following angles is equal to $\widehat{BAC}$?

A $\widehat{ACB}$ **B** $\widehat{ACD}$ **C** $\widehat{ADB}$
D $\widehat{CAD}$ **E** $\widehat{CBD}$

- ③ In Fig. R6, O is the centre of the circle and $\widehat{ABC} = 140^\circ$. Find y .

A 20° **B** 70° **C** 80° **D** 240° **E** 280°

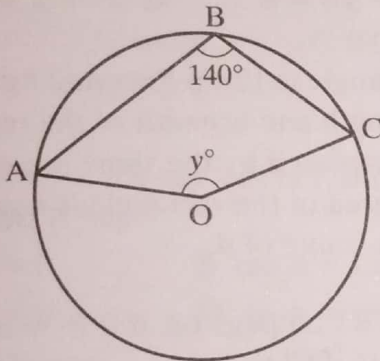


Fig. R6

- ④ The sum of the angles of an n -sided convex polygon is double the sum of the exterior angles. Calculate n .

A 4 **B** 5 **C** 6 **D** 7 **E** 8

- ⑤ In Fig. R7, O is the centre of the circle, $\widehat{WXY} = 80^\circ$ and $\widehat{WXZ} = 45^\circ$. Calculate $\widehat{YWZ}$.

A 35° **B** 45° **C** 55° **D** 80° **E** 125°

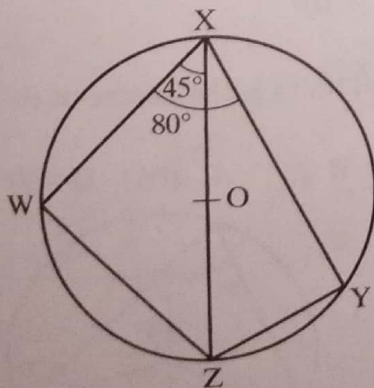


Fig. R7

- ⑥ In quadrilateral WXYZ, XZ bisects $\widehat{WXY}$ and $\widehat{WZY}$. Prove that $\widehat{XWZ} = \widehat{XYZ}$.

- ⑦ In Fig. R8, calculate $|XY|$ and $|CY|$.

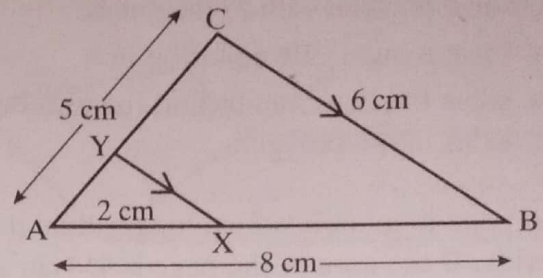


Fig. R8

- ⑧ In quadrilateral ABCD, the lines bisecting $\widehat{ABC}$ and $\widehat{BCD}$ meet at P. Prove that $\widehat{BAD} + \widehat{ADC} = 2\widehat{BPC}$.

- ⑨ In Fig. R9, obtain an equation in x . Hence find the value of x .

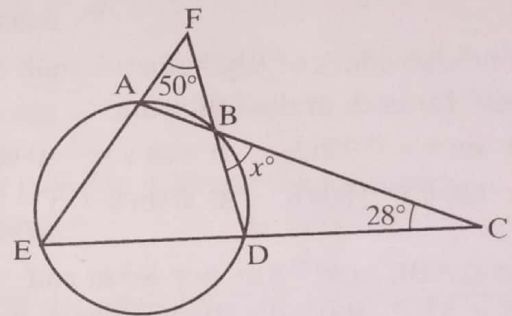


Fig. R9

- ⑩ In $\triangle PQR$, $|PQ| = 6$ cm, $|QR| = 8$ cm, $|RP| = 10$ cm. The bisector of $\widehat{P}$ meets $\overline{QR}$ at N and M is the mid-point of $\overline{QR}$. Calculate $|MN|$.

Revision exercise 3 (Chapters 3, 5)

- ① Solve the following.

a $(a + 6)(a - \frac{1}{2}) = 0$ **b** $3b^2 = 8b$
c $c^2 - 7c - 18 = 0$ **d** $3d^2 = 2d + 5$

- ② Solve the following, leaving a square root sign in the answer if necessary.

a $(x - 3)^2 = 25$ **b** $(x + 7)^2 = 1$
c $(x - 2)^2 = 13$ **d** $2 = (x + 11)^2$

- ③ Solve the equation $2x^2 + 6x + 1 = 0$,

a by completing the square
b by formula.

- ④ Solve the following equations, giving each answer correct to 2 d.p.

a $p^2 - 9p + 5 = 0$ **b** $q^2 = 8 - 5q$
c $3r^2 - r - 7 = 0$ **d** $3 = 8s - 2s^2$

- ⑤ Any n -sided polygon has $\frac{1}{2}n(n-3)$ diagonals. Given a polygon with 27 diagonals,
- form a quadratic equation in n
 - solve the equation to find the number of sides in the polygon.

- ⑥ A page in a book is 5 cm longer than it is wide. If the area of the page is 414 cm^2 , find its length.

- ⑦ Use tables to find the values of the following.

- | | |
|----------------------|-----------------------|
| a $\sin 117^\circ$ | b $\cos 95^\circ$ |
| c $\tan 138^\circ$ | d $\sin 224^\circ$ |
| e $\cos 267.2^\circ$ | f $\tan 196^\circ 6'$ |

- ⑧ Find the values of θ lying between 0° and 360° for each of the following.

- | | |
|--------------------------|---------------------------|
| a $\sin \theta = 0.9563$ | b $\cos \theta = -0.6018$ |
| c $\tan \theta = 0.8098$ | d $6 \sin \theta + 5 = 0$ |

- ⑨ In $\triangle ABC$, $a = 7.8 \text{ m}$, $b = 8.5 \text{ m}$ and $B = 57.7^\circ$. Solve the triangle completely, giving final answers correct to 1 d.p.

- ⑩ Villages P, Q and R are situated as follows. P is 15 km due north of Q and the bearings of P and Q from R are 50° and 80° respectively. Calculate, to the nearest 0.5 km, the distance of R from a P, b Q.

Revision test 3 (Chapters 3, 5)

- ① The greater of the two roots of the equation $(2x-5)(3x+10)=0$ is
A -50 B -10 C -3 D $2\frac{1}{2}$ E 5
- ② One of the roots of the equation $(n-13)^2=9$ is 10. The other root is
A 4 B 16 C 22 D 68 E 94
- ③ Find the value of k such that x^2+5x+k is a perfect square.
A $2\frac{1}{2}$ B $4\frac{1}{4}$ C $6\frac{1}{4}$ D 25 E 100
- ④ If $0^\circ \leq \theta \leq 180^\circ$ and $\cos \theta = -0.5$, then $\theta =$
A 30° B 60° C 120° D 135° E 150°

- ⑤ In which one of the following cases can two triangles PQR be drawn?

- A $p = 5, q = 9, R = 48^\circ$
B $q = 6, r = 5, R = 50^\circ$
C $p = 5, q = 8, P = 90^\circ$
D $p = 3, q = 4, r = 6$
E $p = 5, Q = 72^\circ, R = 64^\circ$

- ⑥ Solve the following quadratic equations.

- a $w^2 - 4w = 5$ b $x^2 - 10x + 16 = 0$
c $2y^2 - 11y = 21$ d $3z^2 + 14z + 8 = 0$

- ⑦ Find x if

- a $(x-3)^2 = 1\frac{7}{9}$ b $(x-2)^2 = 11$
c $(x+5)^2 = 8$ d $x^2 + x = 1$

- ⑧ A rectangle is 12 cm long and 8 cm broad. The length and breadth of the rectangle are both increased by the same amount, d cm. If the area of the rectangle is now 221 cm^2 , find the value of d .

- ⑨ In Fig. R1 on page 68, if $|PB| = 6.6 \text{ cm}$, calculate $|PC|$ to 1 d.p.

- ⑩ In $\triangle ABC$, $b = 5.2 \text{ cm}$, $c = 4.5 \text{ cm}$ and $C = 35^\circ 30'$. Solve $\triangle ABC$ completely, giving final answers correct to 1 d.p.

General revision test A (Chapters 1-6)

- ① Express 0.000 502 in standard form.
A 5.02×10^{-6} B 5.02×10^{-4}
C 5.02×10^{-3} D 5.02×10^3
E 5.02×10^4
- ② In Fig. R10, O is the centre of the circle. Find x .
A 70° B 75° C 105° D 150° E 190°

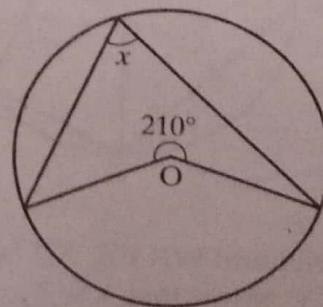


Fig. R10

- ③ If $x^2 - 10x = 24$, then $x = 12$ or
A -3 **B** -2 **C** -1 **D** 1 **E** 2
- ④ If $z^2 + kz + \frac{9}{16}$ is a perfect square, then $k =$
A $\frac{3}{8}$ **B** $\frac{1}{2}$ **C** $\frac{3}{2}$ **D** $\frac{3}{4}$ **E** $\frac{9}{4}$

- ⑤ The population of a city is 2 million to 1 s.f.
 What is the lower limit of the range of population?
A 1 million **B** 1.4 million
C 1.5 million **D** 1.6 million
E 1.9 million

- ⑥ Express 0.028 696 correct to 3 significant figures.
A 0.03 **B** 0.028 **C** 0.029
D 0.028 6 **E** 0.028 7

- ⑦ If $\phi = 248^\circ$, which one of the following statements is false?
A $\tan \phi > 0$ **B** $\tan \phi = \tan 68^\circ$
C $\sin \phi < 0$ **D** $\sin \phi = -\sin 22^\circ$
E $\tan \phi > \sin \phi$

- ⑧ Given the data in Fig. R11, which of the following gives the length of side k ?
A $\frac{8 \sin 35^\circ}{\sin 55^\circ}$ **B** $\frac{8 \sin 55^\circ}{\sin 35^\circ}$
C $\frac{8 \sin 20^\circ}{\sin 55^\circ}$ **D** $\frac{8 \sin 55^\circ}{\sin 20^\circ}$
E None of the above

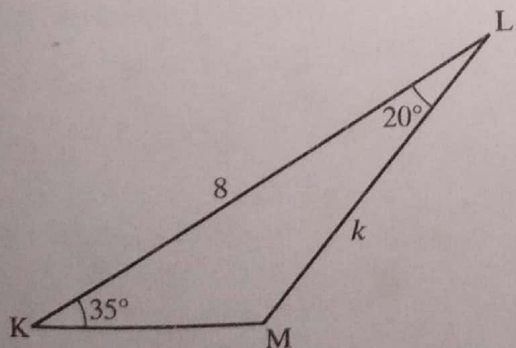


Fig. R11

In Fig. R12 M is the mid-point of PQ. All lengths are in centimetres. Use Fig. R12 to answer questions 9 and 10.

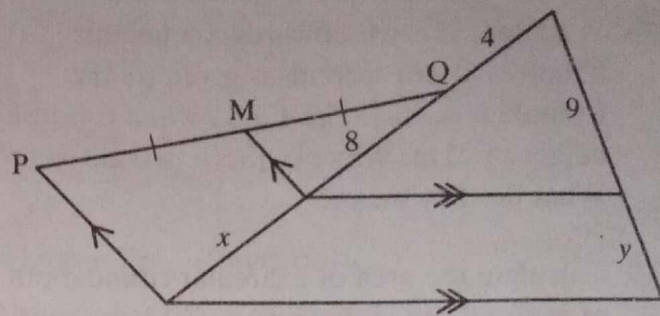


Fig R12

- ⑨ What is the length of the line segment marked x ?
A 3 cm **B** 4 cm **C** 6 cm
D 8 cm **E** 12 cm

- ⑩ What is the length of the line segment marked y ?
A 3 cm **B** 4 cm **C** 6 cm
D 8 cm **E** 12 cm

- ⑪ Use tables to evaluate the following.
a 0.0605^3 **b** $\sqrt[3]{0.0605}$

- ⑫ A regular pentagon ABCDE is inscribed in a circle, centre O. Calculate
a $\angle ABD$ **b** $\angle AOD$.

- ⑬ In Fig. R13, $\overline{AB}$ is a diameter of the circle. Find the value of x .

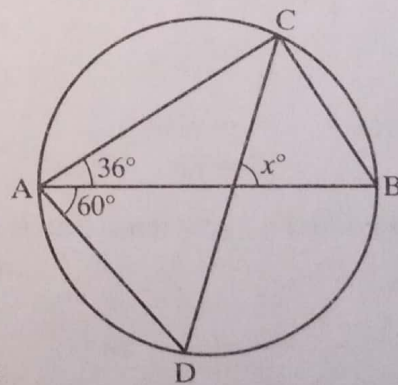


Fig. R13

- ⑭ Solve the following equations, giving answers correct to 1 d.p. where necessary.
a $2x^2 - x - 21 = 0$
b $x^2 + 14x + 8 = 0$
c $2p^2 + 9p - 30 = 0$

- ⑮ A journalist writes n articles and gets ₦200($n + 7$) for each article. If he gets ₦52 000 altogether, how many articles did he write?